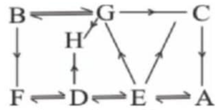
Ans. (b) :

Teacher	Subject				
A	Hindi	English	Mathematics	–	
B	Hindi	English	Geography	History	French
C	–	English	Geography	–	–
D	Hindi	–	Mathematics	–	–
E	–	–	–	History	French

Directions for the following 3 (three) items :
Consider the given information and answer the three items that follow.
Eight railway stations A, B, C, D, E, F, G and H are connected either by two-way passages or one-way passages. One-way passages are from C to A, E to G, B to F, D to H, G to C, E to C and H to G. Two-way passages are between A and E, G and B, F and D, and E and D.

51. While travelling from C to H, which one of the following stations must be passed through?
(a) G (b) E (c) B (d) F

Ans. (b) :



While travelling from C to H, station 'E' must be passed through.
Route $\Rightarrow$ C $\rightarrow$A $\Leftrightarrow$ E $\Leftrightarrow$ D $\rightarrow$H

52. In how many different ways can a train navel from F to A without passing through any station more than once?
(a) 1 (b) 2 (c) 3 (d) 4

Ans. (d) : Route from F to A:-
(i) F $\Leftrightarrow$ D $\rightarrow$ H $\rightarrow$ G $\rightarrow$ C $\rightarrow$ A
(ii) F $\Leftrightarrow$ D $\Leftrightarrow$ E $\Leftrightarrow$ A
(iii) F $\Leftrightarrow$ D $\Leftrightarrow$ E $\rightarrow$ G $\rightarrow$ C $\rightarrow$ A
(iv) F $\Leftrightarrow$ D $\Leftrightarrow$ E $\rightarrow$ C $\rightarrow$ A
Hence, 4 different ways.

53. If the route between G and C is closed, which one of the following stations need not be passed through while travelling from H to C?
(a) E (b) D (c) A (d) B

Ans. (c) : If the route between G and C is closed.
Then,
The route si:-
H $\rightarrow$ G $\Leftrightarrow$ B $\rightarrow$ F $\Leftrightarrow$ D $\Leftrightarrow$ E $\rightarrow$ C
Hence, there is no need to pass through station "A"

GENERAL MENTAL ABILITY

54. If second and fourth Saturdays and ,all the Sundays are taken as only holidays for an office, what would be the minimum number of possible working days of any month of any year?
(a) 23 (b) 22 (c) 21 (d) 20

Ans. (b) :
Let taking month of February of non-leap year
Then, the total number of minimum holidays
= 4 (Sunday) + 2 (Saturday)
= 6
$\therefore$ Minimum working days = 28 – 6
= 22

55. If there is a policy that 1/3rd of population of a community has migrated every year from one place, to some other place, what is the leftover population of that community after the sixth year, if there is no further growth in the population during this period?
(a) 16/243rd part of the population
(b) 32/243rd part of the population
(c) 32/729th part of the population
(d) 64/729th part of the population

Ans. (d) : The leftover population = $=\left(1-\frac{1}{3}\right)^6$

$=\left(\frac{2}{3}\right)^6=\frac{64}{729}$th part

56. 15 students failed in a class of 52. After removing the names of failed students, a merit order list has been prepared in which the position of Ramesh is 22nd from the top. What is his position from the bottom?
(a) 18th (b) 17th
(c) 16th (d) 15th

Ans. (c) :
$\because$ Total number of students = 52
and number of failed student = 15
$\therefore$ Number of passed student = 52 – 15
= 37
Then, Ramesh's position from the bottom
= (37 – 22 + 1)
= (15 + 1)
= 16th

57. There are three pillars X, Y and Z of different heights. Three spiders A, B and C start to climb on these pillars simultaneously. In one chance, A climbs on X by 6 cm but slips down 1 cm. B climbs on Y by 7 cm but slips down 3 cm. C climbs on Z by 6.5 cm but slips down 2 cm. If each of them requires 40 chances to reach the top of the pillars, what is the height of the shortest pillar?
(a) 161 cm (b) 163 cm
(c) 182 cm (d) 210 cm

Ans. (b) :
The height climbed by spider 'A' in 40 Attempts
$\Rightarrow$ 39 (6 – 1) + 6
= 39 × 5 + 6
= 195 + 6
= 201 cm
The height climbed by spider 'B' in 40 attempts
$\Rightarrow$ 39 (7 – 3) + 7
$\Rightarrow$ 39 × 4 + 7
= 156 +7
$\Rightarrow$ 163 cm
The height climbed by spider 'C' in 40 attempts
= 39 (5.5 – 2) + 6.5
= 39 × 4.5 + 6.5
= 175.5 + 6.5
= 182.0 cm
Hence,
The height of shortest pillar is 163 cm.

58. A watch loses 2 minutes in every 24 while another watch gains 2 minutes, in 24 hours. At a particular instant, the two watches showed an identical time. Which of the following statements is correct if 24-hour clock is
(a) The two watches show the identical time again on completion of 30 days.
(b) The two watches show the identical time again on completion of 90 days.
(c) The two watches show the identical time again on completion of 120 days.
(d) None of the above statements correct.

Ans. (d) :
$\because$ A watch loses 2 minutes in every 24 hours while another watch gains 2 minutes in 24 hours.
$\therefore$ The difference in 24 hours $\Rightarrow$ 4 min
The difference in 1 day = 4 min
by option A:- [$\because$ 1 day = 24 hours]
The difference in 30 days = 4 × 30
= 120 min
= 2 hrs
by option B:-
The difference in 90 days = 4 × 90
= 360 min
= 6 hrs
by option C:-
The difference in 120 days = 4 × 120
= 480 min
= 8 hrs
Hence, None of the given statement correct option 'd' is correct.

59. A clock strikes once at 1 o'clock, twice at 2 o'clock and thrice at 3 o'clock, and so on. If it takes 12 seconds to strike at 5 o'clock, what is the time taken by it to strike at 10 o'clock?
(a) 20 seconds (b) 24 seconds
(c) 28 seconds (d) 30 seconds

Ans. (b) :
$\because$ A clock strikes one at 1 o'clock, twice at 2 o'clock and thrice at 3 o'clock and so on.
$\therefore$ The clock strikes 5 times at 5 o'clock.
then, taken time to strike 5 times = 12 sec.
$\therefore$ taken time to strike 10 timer $=\frac{12}{5}\times 10\text{sec}$
= 24 sec
$\because$ The clock strikes 10 times at 10 o'clock.
Hence, The time taken by it to strike at 10 o'clock is 24 second.

60. There are 4 horizontal and 4 vertical lines, parallel and equidistant to one another on a board. What is the maximum number of rectangles and squares that can be formed?
(a) 16 (b) 24 (c) 36 (d) 42

Ans. (c) :
The number of squares = $1^2+2^2+3^2=14$
The number of rectangle = $(1^3+2^3+3^3)-(1^2+2^2+3^2)$
= 36 –14 = 22
The maximum number of rectangles and squares
= 14+22 = 36

61. A bag contains 20 balls. 8 balls are green, 7 are white and 5 are red. What is Hie minimum number of balls that must be picked up from the bag blindfolded (without replacing any of it) to be assured of picking at least one ball of each colour?
(a) 17 (b) 16 (c) 13 (d) 11

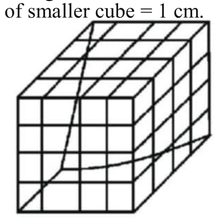
Ans. (b) :
To be assured of picking at least one ball of each colour = 8 + 7 + 1
= 16 balls.

62. If 2 boys and 2 girls are to be arranged in a row so that the girls are not next to each other, how many possible arrangements are there?
(a) 3 (b) 6 (c) 12 (d) 24

Ans. (c) :
All possible arrangements = $\frac{4!}{2!}=\frac{4\times3\times2\times1}{2\times1}=12$

63. The outer surface of a 4 cm x 4 cm x 4 cm cube is painted completely in red. It is sliced parallel to the faces to yield sixty four 1 cm x 1 cm x 1 cm small cubes. How many small cubes do not have painted faces?
(a) 8 (b) 16 (c) 24 (d) 36

Ans. (a) :
$\because$ The length of larger cube = 4 cm.
and the length of smaller cube = 1 cm.

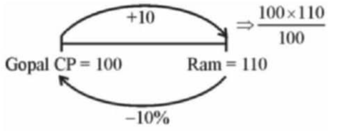


$\because$ The number of cubes which is sliced from one side = 4 (four cut)
$\therefore$ The number of colourless (not painted) cube
= $(x-2)^3=(4-2)^3=(2)^3=8$

General Mathematics/Numerical Ability

64. Gopal bought a cell phone and sold it to Ram at 10% profit. Then Ram wanted to sell it back to Gopal at 10% loss. What will be Gopal's position if he agreed?
(a) Neither loss nor gain (b) Loss 1%
(c) Gain 1% (d) Gain 0.5%

Ans. (c) :
Suppose that Cell phone price = 100 Rs.



$\Rightarrow\frac{100\times110}{100}$

Now Ram sold Cell phone to Gopal
$\frac{110\times90}{100}=99$
Gopal's Profit = 100 – 99 =1%